Stanley Wang

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Experience

Electronic Arts | Commerce & Identity — May 2026 – Aug 2026
Software Engineering Intern — *Redwood City, CA*
- Prototyped a personalized Madden Mobile shopping assistant, projecting $157K in annual revenue lift
- Protected users from duplicate-charge risk through idempotent checkout retries and server-verified purchase recording
- Built reusable agentic commerce components and integration interfaces for expansion across EA's games and web stores
- Built and presented a business case from commerce data and studies, securing approval to advance toward production
- Deployed an issue-to-review agent orchestration project used by teammates on a shared server with per-user Docker workspaces

Autodesk | Media & Entertainment — May 2025 – Aug 2025
Software Engineering Intern — *Montreal, QC*
- Built a diagnostic pipeline combining playback, drawing, and rendering data to debug collaborative film-review sessions
- Built a visual debugger aligning host and participant recordings to pinpoint playback, annotation, and rendering mismatches
- Resolved a production outage for two large studios within an afternoon by correcting feature-gating logic in the login flow

BETA Technologies | Stress & Structures — Jan 2025 – Apr 2025
Software Engineering Intern — *Montreal, QC*
- Automated NASTRAN aircraft model preprocessing in Python, reducing a three-day manual workflow to under 10 seconds
- Developed a reusable Python library for NASTRAN model transformations: DMIG prep, numerical buckling, coordinate symmetry
- Implemented checkpoint-based rollback to generate model variants from shared state without re-parsing or copying

McGill University | AI Ethics Research Lab — May 2024 – Dec 2024
Lead Researcher, supervised by Prof. Joseph Vybihal — *Montreal, QC*
- Wrote a browser extension backend that parsed YouTube video transcripts around most-replayed segments to fact-check claims
- Integrated concurrent Go API clients into Python through CGo and ctypes, handling cross-language memory ownership
- Led prototype architecture and presented the research at McGill's undergraduate research symposium

Education

McGill University, B.Sc. in Computer Science — Sep 2022 – Dec 2026

Skills

Languages: Python, Java, C++, C, Go, SQL, TypeScript, JavaScript, OCaml, Swift, Lua, Bash
Technologies: Linux, Git, Docker, AWS, GitHub Actions, Node.js, React, Flask, FastAPI, REST, Redis, PyTorch, scikit-learn, NGINX

Projects

Mini-C Compiler | Java, MIPS32, JUnit
- Built a Java compiler for a C subset with Pratt parsing, scoped name resolution, type checking, and MIPS32 code generation
- Implemented graph-coloring register allocation with backward liveness analysis, optimistic allocation, and spill-code generation
- Extended the language with classes, single inheritance, and virtual dispatch; built a JUnit/MARS test harness

Operating Systems Projects | C, pthreads
- Built an OS simulator in C with process control blocks, round-robin scheduling, and priority aging
- Simulated demand paging with per-process page tables and LRU page replacement
- Implemented concurrent work dispatch with a producer-consumer queue, using mutexes and semaphores for synchronization

Trading Fours | Python, React, XGBoost, AWS
- Trained an XGBoost genre classifier and combined predictions with Spotify listening history for personalized recommendations
- Deployed containerized Flask and React services to AWS EC2 through GitHub Actions and ECR
- Optimized EC2 memory overhead by sharing model data across workers; added Redis caching and MySQL connection pooling

Pi Talk | Open Source, Agentic Tooling, TypeScript, Node.js, OpenAI TTS
- Published a text-to-speech extension for the Pi agent harness: streaming audio, pause/resume, live speed control
- Implemented Unicode-safe chunking, playback backpressure, timeouts, and cancellation when newer responses arrive